

Career Fair Management System

Final PHP + Oracle 10g XE Project

Windows VM Setup, OCI8 Connection, Run Guide, Demo Plan and Submission Checklist

Web UI	PHP 7.4+ compatible code
Database	Oracle Database 10g XE
Schema	CF_OWNER
PHP-Oracle driver	OCI8
Project URL	http://localhost/career_fair_management/
Connection proof	setup-check.php

Important: This final version does not use MySQL. The PHP website connects directly to the same Oracle 10g XE Career Fair database used for the ADBMS project.

1. What the teacher's bonus requirement means

The shared requirement asks for a working user interface connected to the database created for the course. The documentation must include the earlier interface design screenshots, final implementation/code screenshots, and a detailed explanation of the database connection. A 10-minute video demonstration must show both the user interface and the database connection.

- **BONUS (Up to 5 Marks with Final Term Grade)**

1. User Interface

-You can use any programming language to do this task (Java, C# etc.)

-Make sure to include your midterm interface design screenshots and final term interface design code screenshots in this documentation

2. Database Connection

-The created User Interface must be connected to the database you have created

- Include the detailed description of this process in your report with all necessary screenshots.

****A 10-minute video demo demonstrating the user interface and database connection must be submitted within the deadline via TEAMS as proof. Students are required to create a TEAMS group chat including all group members and the course teacher for submission. The group chat name must follow the format: (ProjectName_Section_Summer2025-2026).**

Important Dates:

1. Project Documentation Submission Deadline (Upload, Softcopy):

20th September 2026, 11:59pm

Submission points from the shared requirement:

- Any programming language may be used for the UI; this package uses PHP.
- Include the midterm interface design screenshots and final implementation/code screenshots in the documentation.
- The UI must be connected to the database created for the project - here, Oracle 10g XE / CF_OWNER.
- Include detailed connection steps and proof screenshots.
- Submit a 10-minute video that demonstrates the UI and the Oracle database connection.
- Create the required TEAMS group/chat including the group members and course teacher, using the required project-name / section / semester naming format.
- The screenshot shows the softcopy deadline as 20 September 2026, 11:59 pm.

2. Final architecture

```
Windows VM
|
+-- Oracle Database 10g XE
|   +-- Schema: CF_OWNER
|   +-- Existing academic tables
|   +-- Existing PL/SQL / triggers / sequences
|   +-- APP_USER (web-login support only)
|
+-- Apache + PHP
    +-- Career Fair PHP Website
        +-- OCI8
            +-- Oracle listener 127.0.0.1:1521/XE
```

The web application does not create a second database. It reads and writes the same Oracle tables used in the project report: CAREER_FAIR, STUDENT, COMPANY, JOB_VACANCY, APPLICATION, INTERVIEW, BOOTH and the other normalized relations.

APP_USER is the only web-support table. It stores three demo login accounts and maps each login to an existing Administrator, Student, or Company Representative ID.

Midterm UI reference

4. User Interface Planning
The interface plan separates the system by user role. The wireframes show a dashboard structure with navigation for career fairs, companies, vacancies, applications, sessions and feedback. During the final term, these designs can be implemented using a programming language and connected to the Oracle database.

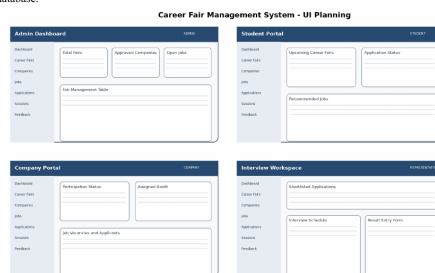


Figure 4. Proposed administrator, student, company and interview interfaces.

Use the original UI-planning screenshot from the report in the final documentation, then place screenshots of the completed PHP pages after it to show the transition from design to implementation.

3. Files included in the final ZIP

Path	Purpose
config/database.php	OCI8 connection to CF_OWNER / XE
diagnostics.php	Shows PHP version, architecture, and whether OCI8 is loaded
setup-check.php	Proves PHP -> OCI8 -> Oracle connection and lists Oracle tables
database/01_web_support_setup.sql	Creates APP_USER and the three demo logins
database/02_verify_project_schema.sql	Checks existing tables, sequences, and counts
database/03_optional_prepare_demo_dates.sql	Moves expired sample dates forward for a live demo
database/99_optional_fresh_schema_ONLY_IF_EMPTY.sql	Creates a fresh schema only for an empty schema
coursework/Career_Fair_36_Queries_Oracle10g.sql	Numbered Oracle coursework queries

Do not run the optional fresh-schema file on your current populated CF_OWNER database. It is only a backup for a completely empty schema.

4. Step 1 - Start and verify Oracle 10g XE

Do all Oracle work inside the Windows VM where Oracle 10g XE is already installed.

- Open Windows Services (Win + R -> services.msc). Start the Oracle XE database and listener services if they are stopped.
- Open the Oracle Database Express Edition home page / APEX and sign in as CF_OWNER.
- Go to SQL -> SQL Commands.

Run these checks one at a time

```
SELECT USER FROM dual;
```

```
SELECT table_name  
FROM user_tables  
ORDER BY table_name;
```

```
SELECT sequence_name  
FROM user_sequences  
ORDER BY sequence_name;
```

The main project should contain the 18 normalized academic tables. If they already exist, keep them. Do not recreate them.

Confirm key data

```
SELECT COUNT(*) FROM student;  
SELECT COUNT(*) FROM career_fair;  
SELECT COUNT(*) FROM company;  
SELECT COUNT(*) FROM job_vacancy;  
SELECT COUNT(*) FROM application;
```

If your data is present, continue to the web-support setup.

5. Step 2 - Add only the web login support table

In Oracle APEX use SQL -> SQL Scripts -> Upload, because the support file contains multiple SQL statements.

- Upload database/01_web_support_setup.sql.
- Run it once while logged in as CF_OWNER.
- It creates APP_USER and SEQ_APP_USER and inserts three demo accounts.
- It does not replace the academic tables.

Demo accounts

Role	Username	Password	related_id
Admin	admin@careerfair.local	Admin@123	Administrator 1
Student	arif@student.aiub.edu	Student@123	Student 1
Company Rep	farhan@technova.com	Company@123	Representative 1

If your current Oracle IDs are different, update APP_USER.related_id so the login points to an existing row.

```
SELECT user_id, username, role, related_id, display_name
FROM app_user
ORDER BY user_id;
```

6. Step 3 - Install Apache/PHP and enable OCI8

Recommended for this legacy Oracle 10g XE setup: keep Apache/PHP and Oracle on the same Windows VM. The generated project is compatible with PHP 7.4 syntax and later.

Compatibility note: the PHP OCI8 documentation states that PHP 7 uses OCI8 2.2.0. Oracle Client 11.2 can connect to Oracle Database 9.2 or later, and Oracle Client 12.1 can connect to Oracle Database 10.2 or later. For an Oracle 10g XE database, use an Oracle Client/Instant Client version that still supports 10g, such as 11.2 or 12.1. Do not use an arbitrary newer client if it drops 10g connectivity.

A. Install/start XAMPP

- Install XAMPP inside the Windows VM.
- Start Apache. MySQL is not required for this final project.
- Open `http://localhost/` and confirm Apache works.

B. Check your PHP build

```
C:\xampp\php\php.exe -v
C:\xampp\php\php.exe -i | findstr /I "Architecture Thread Safety"
```

For Apache module use, choose the Thread Safe (TS) OCI8 DLL that matches your exact PHP version and x86/x64 architecture. The Oracle client libraries must match the PHP architecture.

C. Enable OCI8

- First check `C:\xampp\php\ext` for an OCI8 DLL already supplied with your PHP build.
- If not present, obtain the Windows OCI8 build matching your exact PHP version, TS/NTS mode and x86/x64 architecture.
- Install a compatible Oracle Instant Client (11.2 or 12.1 is appropriate for Oracle 10g connectivity) with the same x86/x64 architecture as PHP.
- Add the Instant Client folder to the Windows PATH. Restart Apache (or restart Windows if PATH changes are not picked up).
- Edit `C:\xampp\php\php.ini` and enable exactly one OCI8 extension DLL. The exact filename depends on the build, commonly `php_oci8_11g.dll` or `php_oci8.dll`.

```
; example only - use the DLL that actually exists in your ext folder
extension=php_oci8_11g.dll
```

D. Verify OCI8

```
C:\xampp\php\php.exe -m | findstr /I oci8
```

You should see **oci8**. If Apache will not start, check `C:\xampp\apache\logs\error.log` and verify that PHP, the OCI8 DLL and Oracle client are the same architecture.

7. Step 4 - Copy and configure the project

Extract the ZIP and place the project folder here:

```
C:\xampp\htdocs\career_fair_management
```

Keep that exact folder name for the easiest setup. If you rename it, update BASE_URL in config/app.php.

Edit Oracle credentials

Open:

```
C:\xampp\htdocs\career_fair_management\config\database.php
```

Change only the password first:

```
define('DB_USER', 'CF_OWNER');  
define('DB_PASS', 'YOUR_REAL_CF_OWNER_PASSWORD');  
define('DB_CONNECTION_STRING', '127.0.0.1:1521/XE');
```

Do not change the service from XE unless your Oracle listener uses a different service name or port.

Check PHP/OCI8 before the database login

```
http://localhost/career_fair_management/diagnostics.php
```

Expected result: **OCI8 is loaded: oci_connect() is available.**

8. Step 5 - Prove the Oracle connection

`http://localhost/career_fair_management/setup-check.php`

A successful page should show:

- Oracle connection successful.
- Oracle schema = CF_OWNER.
- Oracle server version.
- A list of the project tables including APP_USER.

Take a full screenshot of this page with the Windows date/time visible. This is one of the strongest screenshots for the Database Connection section of the final documentation.

If connection fails

Error	What to check
Call to undefined function oci_connect()	OCI8 is not loaded; fix php.ini/DLL/client and restart Apache.
ORA-01017	Wrong CF_OWNER username/password.
ORA-12154	Oracle service name / naming problem; use Easy Connect 127.0.0.1:1521/XE.
ORA-12541	Listener is not running.
ORA-12514	Listener is running but XE service is not registered / wrong service.
Apache will not start	Architecture/DLL clash. Check Apache error.log and PATH.

9. Step 6 - Login and test each portal

Admin

Login: admin@careerfair.local / Admin@123

- Dashboard: counts from Oracle.
- Career Fairs: create/edit/delete using SEQ_CAREER_FAIR and Oracle DATE values.
- Companies: approve or reject fair participation.
- Booths: add, assign and release booths using an Oracle transaction.
- Jobs, Applications, Interviews, Sessions and Feedback: read live Oracle data.

Student

Login: arif@student.aiub.edu / Student@123

- Register for a fair (Oracle FAIR_REGISTRATION).
- Browse OPEN job vacancies and apply (Oracle APPLICATION + SEQ_APPLICATION).
- View application history.
- Register for fair sessions.
- Submit feedback (Oracle FEEDBACK + SEQ_FEEDBACK).

Company representative

Login: farhan@technova.com / Company@123

- Request career-fair participation.
- Create/edit vacancies (Oracle JOB_VACANCY + SEQ_JOB).
- Review and shortlist applications.
- Schedule interviews only for SHORTLISTED applications, matching the business rule.

If your sample fair/job dates are already past, run database/03_optional_prepare_demo_dates.sql once before recording the demo.

10. Step 7 - Demonstrate that the UI really changes Oracle

For the video and screenshots, do not only show the website. Perform a change in PHP, then immediately prove the same change in Oracle SQL Commands.

Example A - Company approval

1) In Admin -> Companies, change a participation request to APPROVED.

2) In Oracle SQL Commands run:

```
SELECT company_id, fair_id, approval_status
FROM company_participation
ORDER BY fair_id, company_id;
```

Example B - Student application

1) Login as Student and apply for an OPEN job.

2) In Oracle run:

```
SELECT application_id, student_id, job_id,
       application_date, application_status
FROM application
ORDER BY application_id DESC;
```

Example C - Booth assignment

1) Admin -> Booths -> assign an AVAILABLE booth to a company approved for that fair.

2) In Oracle run:

```
SELECT ba.booth_id, ba.company_id, ba.assignment_date,
       b.booth_number, b.booth_status
FROM booth_assignment ba
JOIN booth b ON b.booth_id = ba.booth_id
ORDER BY ba.booth_id;
```

This PHP action -> Oracle SELECT proof is what makes the database connection obvious to the teacher.

11. Recommended 10-minute demonstration script

Time	What to show
0:00-0:45	Introduce project, PHP UI, Oracle 10g XE, CF_OWNER.
0:45-1:30	Oracle APEX: SELECT USER FROM dual and show project tables.
1:30-2:15	Open diagnostics.php, then setup-check.php showing successful OCI8/Oracle connection.
2:15-3:15	Admin login and dashboard; explain counts come from Oracle.
3:15-4:20	Admin: approve a company or edit a fair, then prove it in Oracle SQL Commands.
4:20-5:30	Admin: booth transaction or session creation; prove row in Oracle.
5:30-6:45	Student login: register/apply; prove APPLICATION or FAIR_REGISTRATION in Oracle.
6:45-8:00	Company login: review/shortlist application and show job/interview pages.
8:00-9:00	Show config/database.php connection code and explain OCI8 / XE service.
9:00-10:00	Final Oracle SELECT proof, recap tables/UI/connection and finish.

Keep the Windows date/time visible in important evidence shots when possible, especially if your course requires timestamped Oracle screenshots.

12. Screenshots to include in the final documentation

- Teacher's final-term UI/database connection requirement screenshot.
- Midterm UI planning screenshot from the report.
- Final login page.
- PHP/OCI8 diagnostics page showing OCI8 loaded.
- Oracle Connection Check page showing CF_OWNER and detected tables.
- config/database.php connection code (hide or blur the real password).
- Admin dashboard.
- Career Fair management page.
- Company approval page.
- Booth management page.
- Student Career Fair / Jobs page.
- Company Applications / Interviews page.
- At least one UI operation followed by the matching Oracle SELECT result.
- Any final UI code screenshot requested by the teacher.

Never show the real CF_OWNER password in the report or video. Replace it with ***** or crop that line.

13. Existing Oracle triggers and constraints

Your Oracle project already contains business rules. The PHP website does not bypass them; Oracle remains the final authority. If an action is rejected, the website shows the Oracle error message.

Situation	Meaning / fix
Registration deadline error	The fair deadline is in the past. Use a future fair or run the optional demo-date script.
CK_FAIR_REG_DEADLINE violated	Registration deadline must satisfy the fair date constraint. Update event date and deadline together.
Only an AVAILABLE booth can be assigned	Choose a booth whose BOOTH_STATUS is AVAILABLE.
Company approval error during booth assignment	The selected company must be APPROVED for the same fair as the booth.
Only shortlisted applicants can be interviewed	Set APPLICATION_STATUS to SHORTLISTED before scheduling the interview.
Duplicate registration/application	The same composite/unique relationship already exists; use another record.

14. Final pre-submission checklist

- Oracle 10g XE and listener start correctly in the Windows VM.
- CF_OWNER can log in and all academic tables are present.
- APP_USER exists and the three demo logins work.
- diagnostics.php reports OCI8 loaded.
- setup-check.php reports Oracle connection successful.
- Admin, Student and Company portals all open without PHP errors.
- At least one CREATE/UPDATE action from PHP is verified by an Oracle SELECT.
- No MySQL/phpMyAdmin screenshots are used for the final database connection proof.
- Midterm UI design screenshot is included in documentation.
- Final UI screenshots and connection-code screenshots are included.
- Real Oracle password is hidden in screenshots.
- 10-minute demo video shows both UI and Oracle.
- Required TEAMS group/chat is created with the requested naming format.
- Softcopy is uploaded before the stated deadline.

Recommended final folder

```
Final_Submission/  
|-- Project_Report.pdf  
|-- Career_Fair_Management_PHP_Oracle10g_Final_Project.zip  
|-- 10_Minute_Demo.mp4  
|-- Screenshots/  
|   |-- Connection_Check.png  
|   |-- Admin_Dashboard.png  
|   |-- UI_to_Oracle_Proof.png  
|   |-- ...
```

Technical source notes

OCI8 setup guidance in this document follows the PHP OCI8 manual and PECL OCI8 package notes: PHP 7 uses OCI8 2.2.0; Windows builds must match PHP version, Thread Safe/Non-Thread Safe mode and x86/x64; Instant Client must be on PATH; Oracle Client 11.2 supports connections to Oracle Database 9.2 or later, while Client 12.1 supports Oracle Database 10.2 or later.

The generated project has been PHP syntax-checked. Because this generation environment does not contain your Windows Oracle 10g XE instance, the final runtime test must be performed in your VM using diagnostics.php and setup-check.php.